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MULTI-DOOR SAFETY INTERLOCK SYSTEM

A Smart Vehicle Safety Solution

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OUR SQUAD

Team Members

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ABOUT THE PROJECT

What does it do?

Project Overview

The Multi-Door Safety Interlock System is a purely hardware-based vehicle safety solution. It prevents the vehicle engine from starting unless ALL four doors are fully closed and latched.

At its heart lies the 74HC08 Quad AND Gate IC. The four door sensors (represented by DIP switches) feed into a chain of AND gates. The engine enable signal (LED) goes HIGH only when every single input is HIGH.

The moment any door opens, its switch goes LOW, the AND chain output drops to LOW, and the engine is immediately disabled — a true passive, fail-safe hardware interlock.



4-Door Interlock

ALL doors must be closed for engine to start



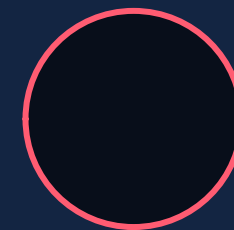
AND Gate Logic

74HC08 chains 4 AND gates — pure hardware logic



Fail-Safe Design

Any open door immediately disables the engine



LED Output Signal

Bright LED signals engine-ready state

COMPONENTS USED

Hardware components and their function in the system

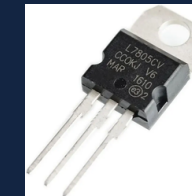
9V Battery

Main power supply for the entire circuit



5V Regulator (7805)

Regulates 9V down to stable 5V for the IC



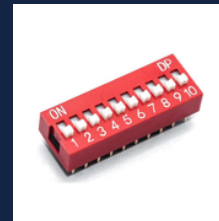
74HC08 IC

Quad AND gate — the logic core of interlock



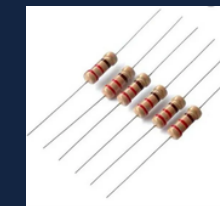
DIP Switch (4-pos)

Simulates 4 door open/close sensors



Resistors

Pull-down & current-limiting resistors



LED

Output indicator: ON = all doors closed



Breadboard

Solderless prototyping platform



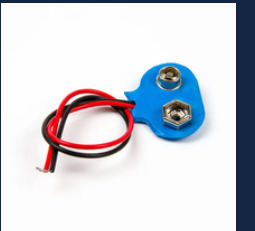
Jumper Wires

Connections between all components



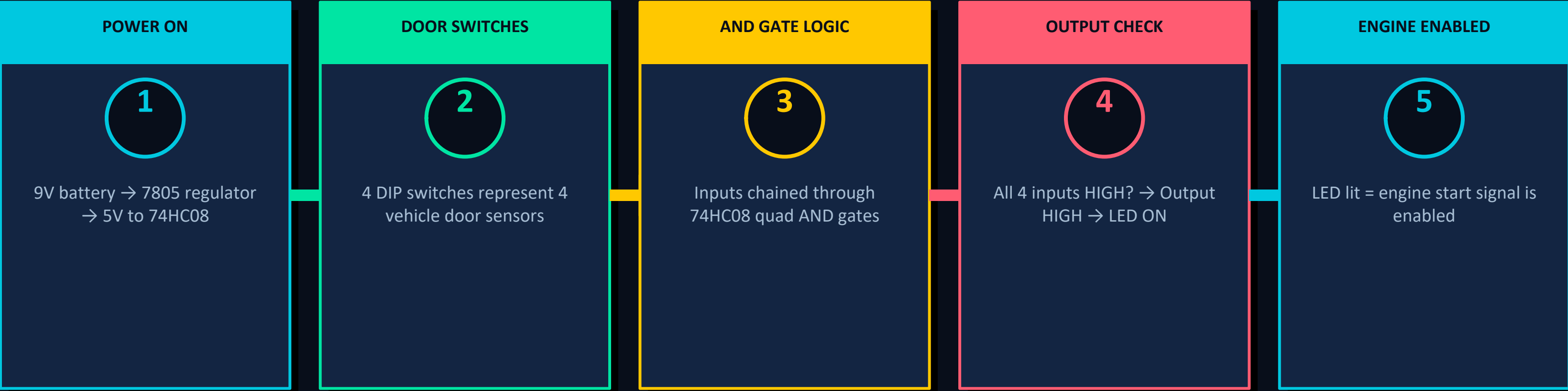
Battery Connector

Connects 9V battery to the circuit safely



HOW IT WORKS

Logic flow of the Multi-Door Safety Interlock System

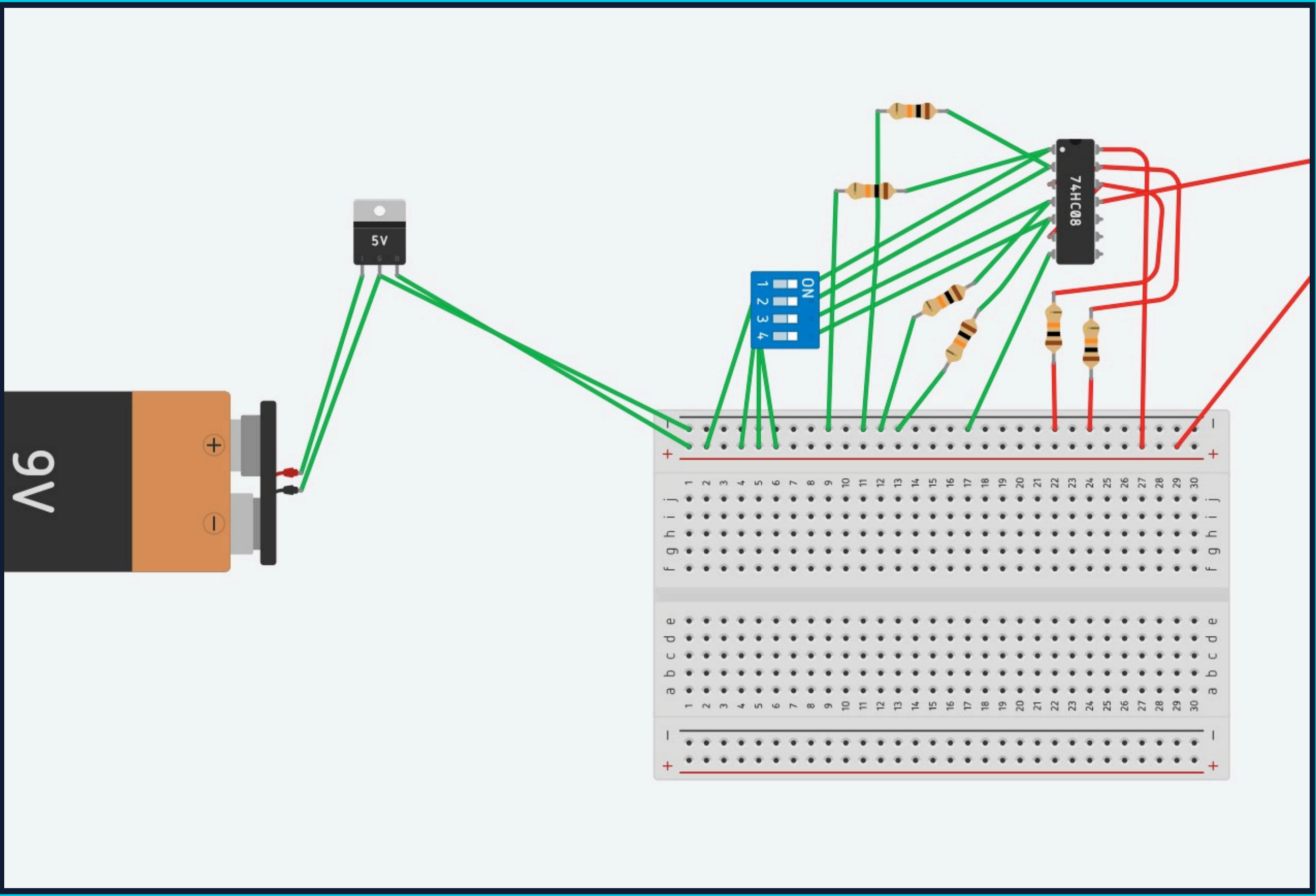


TRUTH TABLE SUMMARY

Door 1	Door 2	Door 3	Door 4	OUTPUT
CLOSED	CLOSED	CLOSED	CLOSED	HIGH → Engine ON
OPEN	CLOSED	CLOSED	CLOSED	LOW → Engine OFF
CLOSED	OPEN	CLOSED	OPEN	LOW → Engine OFF

CIRCUIT DIAGRAM

Breadboard layout — Multi-Door Safety Interlock



Power

9V → 7805 → 5V DC

Logic

4× AND gates (74HC08)

Input

4-pos DIP switch

Output

LED = Engine enable

PROJECT EXPENSES

Estimated component cost breakdown (INR)

Component	Min (₹)	Max (₹)
9V Battery	₹30	₹50
Battery Connector	₹10	₹15
5V Voltage Reg (7805)	₹15	₹25
Breadboard	₹80	₹150
74HC08 IC	₹20	₹40
DIP Switch (4-pos)	₹20	₹40
Resistors	₹10	₹20
LED	₹5	₹10
Jumper Wires	₹30	₹50
TOTAL ESTIMATED COST	₹220	₹400

BUDGET

₹220

to

₹400

Very Affordable!

Low-cost, high-impact safety solution for vehicles of all types.



THANK YOU